# Chang Liu

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## Research Interests

Combinatorial Optimization • Machine Learning • Computational Mathematics
Integer Programming • Constraint Programming • Search Algorithms

## Education

**University of Toronto**
Ph.D. Candidate, Operations Research and Machine Learning 2022 - present

Supervisor: Dr. Elias B. Khalil

Research Topic: *"Improving Search in Combinatorial Optimization Through Machine Learning"*. The research proposes to improve combinatorial optimization problems solving by studying branching heuristics, more specifically through the discovery of backdoor variables. We seek to contribute through both the theoretical perspectives as well as proposing concrete algorithmic improvements.

**University of Toronto**
M.A.Sc., Operations Research 2018

Supervisors: Dr. J. Christopher Beck, Dr. Dionne M. Aleman

Dissertation: *"Modelling and Solving the Senior Transportation Problem"*. The thesis presents mathematical optimization models to solve a deterministic and a stochastic version of the combinatorial optimization problem - the Senior Transportation Problem, which is shown to be $\mathcal{NP}$-hard. Extensive empirical experiments have been conducted and an analysis of the performance and efficiency of each algorithm is presented. [pdf]

**University of Waterloo**
B.Math., Combinatorics and Optimization, Spanish Minor 2014

**Wilfrid Laurier University**
B.BA., Supply Chain Management 2014

## Publications

**Refereed Journal Publication**

[1] Morin, M., Castro, M.P., Booth, K.E.C., Tran, T.T., **Liu, C.**, & Beck, J.C. "Intruder Alert! Optimization Models for Solving the Mobile Robot Graph-Clear Problem", *Constraints, 23(3), 335-354*, 2018. **Winner of the Distinguished Paper Award at CPAIOR 2018.** [pdf]

**Refereed Conference Publications**

[2] Deza, A., **Liu, C.**, Vaezipoor, P., & Khalil, E.B., "Fast Matrix Multiplication Without Tears: A Constraint Programming Approach", *Proceedings of the 29th International Conference on Principles and Practice of Constraint Programming, 14:1–14:15*, 2023. [pdf]

[3] **Liu, C.**, Aleman, D.M., & Beck, J.C., "Modelling and Solving the Senior Transportation Problem", *Proceedings of the 15th International Conference of Constraint Programming, Artificial Intelligence, and Operations Research, 412-428*, 2018. [pdf]

[4] Rosu, D., Aleman, D.M., Beck, J.C., Chignell, M., Consens, M., Fox, M.S., Gruninger, M., **Liu, C.**, Ru, Y., & Sanner, S., "A Virtual Marketplace for Goods and Services for People with Social Needs", *Proceedings of the IEEE Canada International Humanitarian Technology Conference 2017, 201-206*, 2017. [pdf]

### Refereed Workshop Publications

[5] **Liu, C.**, Budhkar, A., "Data-driven Optimization Model for Global Covid-19 Intervention Plan", *ICLR 2021 Workshop: Machine Learning for Preventing and Combating Pandemics*, 2021. [pdf]

[6] Du, E.S.J., **Liu, C.**, Wayne, D.H. "Automated Fashion Size Normalization", *13th ACM Conference on Recommender Systems, Workshop on Recommender Systems in Fashion*, 2019. [pdf]

[7] Collet, S., Dadashi, R., Naram, Z., **Liu, C.**, Sobhani, P., Vahlis, Y., & Zhang, J.C. "Enabling Data Aggregation Using Differential Privacy", *CCS Workshop on Theory and Practice of Differential Privacy*, 2018.

[8] Collet, S., Dadashi, R., Naram, Z., **Liu, C.**, Sobhani, P., Vahlis, Y., & Zhang, J.C. "Boosting Model Performance through Differentially Private Model Aggregation", *NeurIPS Women in Machine Learning Workshop*, 2018. [pdf]

[9] **Liu, C.**, Aleman, D.M., & Beck, J.C., "The Senior Transportation Problem", *AAAI Workshop on AI and OR for Social Good*, 2017. [pdf]

[10] Rosu, D., Aleman, D.M., Beck, J.C., Chignell, M., Consens, M., Fox, M.S., Gruninger, M., **Liu, C.**, Ru, Y., & Sanner, S., "Knowledge-Based Provision of Goods and Services for People with Social Needs: Towards a Virtual Marketplace, *AAAI Workshop on AI and OR for Social Good*, 2017. [pdf]

## Teaching Experience

### Course Lecturer

- MIE 562: *Scheduling*, 2024

### Teaching Assistant

- MIE 524: *Data Mining*, 2023
- MIE 376: *Mathematical Programming (Optimization)*, 2023
- MIE 262: *Operations Research I*, 2023, 2017.
- MIE 562: *Scheduling*, 2023, 2022.
- MIE 376: *Cases in Operations Research*, 2017.
- MIE 1620: *Linear Programming and Network Flow Theory*, 2016.

## Industry Experiences

**Applied Research Scientist** 2017 - 2022
R&D Team, Georgian Partners, Toronto

RESPONSIBILITIES
- Engage in applied research projects with AI startups in areas including NLP representation learning, constrained optimization, recommender systems and differential privacy
- Consult portfolio companies on technologies and best practices in machine learning and optimization
- Create technical content including papers, technical reports, and blog posts

SELECTED PROJECTS

*Shipment Assembly Optimization*
- Built an optimization framework that combines pickup and delivery orders into a minimal number of total shipments while minimizing travel costs.
- Investigated Benders decomposition, integer programming, and heuristical approaches.
- Reduced total travel time of 200 orders that would normally take 11,000 hours down to 4000 hours.

*Fashion Sizing Recommender System* [5]
- Learn users clothing fit preferences through historical purchase data and predict sizing for other garments across different categories and brands
- Developed a quadratic programming approach to solve the optimization problem with a 300x speed up from traditional gradient descent based methods

*Creating a Fair and Healthy Online Marketplace*
- Designed and implemented an optimization framework on top of a recommender system to balance supply and demand on the marketplace
- Modelled problem using mixed integer programming approaches
- 25% lift in user retention and engagement

*Machine Learning Model Aggregation through Differential Privacy* [6], [7]
- Implemented differentially private logistic regression and support vector machine with MXNet
- Boosted average model performance by 9.72% through private model aggregation and shortened model adoption time from months to days
- Code developed is then transferred to TensorFlow and contributed to TensorFlow/privacy

**Risk Analyst** 2014 - 2015
Manulife Bank, Manulife Financial Group, Waterloo
- Responsible for credit risk analysis of the Bank's lending portfolio
- Conduct research on current events related to the Bank's risk exposure

## Professional Activities

**Research Presentations & Invited Talks**

- Building differentially private machine learning models using TensorFlow, *O'Reilly AI Conference London*, Oct 2019.
- Solving the cold start problem: Data and model aggregation using differential privacy, *Strata Data Conference New York*, Sept 2018.
- Where AI and Business Converge, *Toronto Women's Data Group*, Aug 2018.
- Modelling and Solving the Senior Transportation Problem, *Sixteenth International Conference on the Integration of Constraint Programming, Artificial Intelligence, and Operations Research*, June 2018.
- AI Trust, *AI and How It Will Affect The World We Live In*, June 2018.
- How to Implement AI in your Software Solution, *Machine Intelligence Toronto*, Nov 2017.
- Exact Methods for the Senior Transportation Problem. *Annual Institute of Industrial & Systems Engineers Conference and EXPO*, May 2017.
- The Senior Transportation Problem. *AAAI Workshop on AI and OR for Social Good*, Feb 2017.

**Interviews & Podcasts**

- Epsilon Software for Private Machine Learning with Chang Liu, *This Week in Machine Learning & AI*, May 2018.
- How privacy-preserving techniques can lead to more robust machine learning models, *The O'Reilly Data Show Podcast*, August 2018.
- Understanding Differential Privacy with Chang Liu, *The Georgian Impact Podcast — AI, ML & More*, August 2018.

## Awards and Scholarships

**First Place for Oral Presentation** 2017
University of Toronto

**University of Toronto Fellowship** 2015 - 2017
Industrial Engineering

**William D. Packham Undergraduate Entrance Scholarship** 2008
University of Waterloo

**University of Waterloo President's Scholarship** 2008
University of Waterloo

**René Descartes Scholarship** 2008
University of Waterloo

## Technical Skills

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| Programming | Python, C++, Java, SQL, R, PySpark |
| Python Libraries | Tensorflow, MXNet, Keras, ScikitLearn, Pandas, Numpy |
| Optimization | CPLEX, CP Optimizer, Gurobi, OR-Tools, AMPL |
| Cloud Services | AWS, GCP |